

WENXIN LIU

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Education

University of Warwick

Doctor of Philosophy in Applied Mathematics

Sep. 2026 – Sep. 2030

Imperial College London

Master of Science in Applied Mathematics (Scientific computing and Machine Learning)

Sep. 2024 – Nov. 2025

First class Distinction

Xi'an Jiaotong Liverpool University

Bachelor of Science in Applied Mathematics

Sep. 2020 – Jun. 2024

First class honors

Highly relevant experiences with numerical analysis and optimization

Research paper at Imperial College London

Oct. 2025–May. 2026

- Paper accepted by European Control Conference 2026 coauthor with Dr.Dante Kalise: <http://arxiv.org/abs/2511.14355>: A PDE-constrained Optimization Approach to Optimal Trajectory Planning under Uncertainty via Reflected Schrödinger Bridges.
- Paper accepted by International Symposium on Mathematical Theory of Networks and Systems(MTNS) 2026 coauthor with Dr.Daisuke Inoue and Dr.Dante Kalise: Computational Methods for Generalized Schrödinger Bridge Problems.

Research at National supercomputing center

Sep. 2025–Mar. 2026

- Developed a hybrid **GNN–MLP** framework for intelligent solver selection in sparse linear algebra. Learned graph-based structural representations together with matrix-level features to automatically determine optimal preconditioner and iterative solver combinations. Demonstrated more than **3.75× acceleration** in solving large-scale sparse linear systems arising from scientific computing applications.
- Developed a machine learning-based optimization framework for **Algebraic Multigrid (AMG)** solvers, enabling automatic tuning of critical parameters such as the strong connection threshold θ . Improved convergence efficiency and solver robustness for challenging physical simulation workloads.
- Developed a **GNN-based initial guess prediction** framework for iterative linear solvers by reformulating the initialization problem as a graph representation learning task. Represented sparse matrices as graphs, extracted structural features through a graph encoder, and generated adaptive initial solution vectors using an attention-based decoder. Achieved an average **1.79×** and maximum **19.51×** reduction in solution time for challenging pressure Poisson systems.

Postgraduate Final Project about Optimal Transport

Jan. 2025–Sep. 2025

- Conduct symmetric Douglas-Rachford splitting and prime-dual method with proximal operator for solving the dynamic optimal transport problem, compare its performance with the Sinkhorn algorithm.
- Use Schrödinger Bridge to approximate the solution of optimal transport problems in both dynamic way and static way, and implement the algorithm for solving the Schrödinger Bridge problem with Forward and Backward system with proximal recursion.
- Implement the Schrödinger Bridge problem with nonlinear prior drift and reflective boundary conditions, and apply the proximal recursion algorithm and Finite Element Method to solve the problem with prior drift and boundary conditions like obstacles in both 2D and 3D cases.

Computational Partial Difference Equation Project

Jan. 2025 – Apr. 2025

- Solve second order Nonlinear ODE and nonlinear heat equation with explicit method, lagged approach and implicit Newton linearization, analyze numerical result with physical meaning and theoretical perturbation analysis.
- Develop numerical solutions for elliptic PDEs with non-uniform boundary conditions using structured grid mapping and adaptive mesh refinement (AMR) to enhance accuracy in complex geometries. Implement iterative solvers (Gauss-Seidel, SOR, multigrid) for large sparse linear systems.
- Solve wave equation with Neumann and transparent boundary conditions and perform discrete Fourier analysis to assess numerical dissipation and dispersion. Design a position-dependent Perfectly Matched Layer (PML) to reduce boundary reflections by 85%. Validated accuracy and convergence through analytical Fourier analysis.
- Design an IMEX finite difference framework for coupled wave-heat systems, combining explicit wave solvers and implicit thermal diffusion. Derived stability criteria via Fourier and Jury analysis, and implement efficient solvers using staggered grids. Demonstrated the effect of thermal diffusivity on energy dissipation and wave behavior with vary of variables through numerical experiments.

Finite Element: theory and implementation Project

Jan. 2025 – Apr. 2025

- Implement an object-oriented finite element framework in Python to solve elliptic PDEs (Helmholtz/Poisson equations), featuring modular classes for Lagrange elements, function spaces, structured meshes (UnitSquareMesh), and Gauss quadrature rules. Engineer a global node numbering system to map local-to-global degrees of freedom, enabling efficient sparse matrix assembly via cell-wise contributions. Utilized einsum-optimized tensor operations to compute stiffness/mass matrices.

- Engineer a mixed finite element solver for Stokes equations ($\mu \Delta \mathbf{u} = \nabla p + \mathbf{f}, \nabla \cdot \mathbf{u} = 0$) with structured 2D meshes. Design vector-valued Lagrange elements with tensor-product basis functions, extending scalar tabulation to handle 2D velocity fields via Vandermonde matrix-based evaluations. Implemented divergence and gradient coupling terms through block matrix assembly ($A = \begin{bmatrix} A_u & B^T \\ B & 0 \end{bmatrix}$) using SciPy's sparse lil_matrix format, and stabilized the saddle-point system via pressure Poisson stabilization (diagonal perturbation). Optimize strain-rate tensor ($\epsilon(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$) integration using einsum -accelerated operations. Then deploy the solver to simulate incompressible viscous flows ($\mu = 1$) with analytic benchmarks derived from stream functions ($\psi(x, y) = (1 - \cos 2\pi x)(1 - \cos 2\pi y)$).

Data Science Project

Jan. 2025 – Apr. 2025

- Develop multi-class classifiers (Decision Trees, Random Forests) and Huberised SVM models for airfield weather condition prediction. Optimized Random Forests via Gini importance-based feature selection. Stabilized SVM training with modified Huber loss and mini-batch SGD, achieving 94.1% test accuracy.
- Design a compound loss MLP for joint runway temperature prediction with physical constraints. Implemented transfer learning by freezing pretrained MLP hidden layers and fine-tuning sigmoid output layers for binary classification, achieving accuracy to 99.8% versus 97% for non-transferred models.
- Design a U-Net convolutional neural network in PyTorch for joint dimensionality reduction and gene expression regression. Achieved $R^2 = 0.69$ on regression with early stopping and dropout.
- Construct a spectral layout using normalized Laplacian eigenvectors for data analysis and clustering, conducting analysis for degree centrality via bipartite histograms.

Computational Linear Algebra Project

Sep. 2024–Jan. 2025

- Develop QR-based solvers for least squares problems (over/underdetermined systems), implementing 4 algorithms: Normal Equations, Classical/Modified Gram-Schmidt, and Householder (Python/Numpy). Design regularization method for ill-posed least-square problems using augmented QR factorization, reducing condition number by 5 orders of magnitude with $\lambda = 1e - 6$ penalty.
- Implement and analyze the practical QR algorithm for computing eigenvalues of real symmetric matrices, including enhancements such as Wilkinson shift and deflation. Demonstrate the efficiency of Wilkinson shift in accelerating convergence through comparative experiments across varying matrix sizes, and verify eigenvalue accuracy via inverse iteration.
- Extend the GMRES algorithm by implementing Jacobi and Gauss-Seidel preconditioners to improve convergence on ill-conditioned systems. Conduct a comparative analysis of residual decay and convergence rates for varying ϵ values, showing that preconditioning significantly improves performance, especially with Gauss-Seidel under increasing ill-conditioning.

Scientific computing Project

Sep. 2024 – Jan. 2025

- Implement and analyze linear search, merge sort with binary search method ($O((m + n) \log n)$) for searching and pattern matching in large datasets. Apply the Rabin-Karp algorithm for pattern matching in amino acid sequences, utilizing optimized hash calculations for fast sequence comparison.
- Develop and analyze numerical solutions for a nonlinear dynamical system of ODEs. Implemented tests comparing perturbation energy calculations with simulations, showing consistency and accuracy. Analyzed periodic, aperiodic, and chaotic behaviors for different system sizes, using Principal Component Analysis (PCA) and correlation dimension analysis to identify chaotic dynamics.
- Implement the extended Kalman filter for Lorenz system, analyze the computational cost of one step of the extended Kalman filter. Focused on identifying the most computationally expensive portions of the method and explored trends in wall time with varying problem sizes.

Undergraduate Final Project about Numerical Optimization

Sep. 2023–Jun. 2024

- Apply the interior point method by transforming inequality-constrained problems into equality-constrained forms using indicator functions, enabling the use of barrier methods for numerical solution approximation, and implement the algorithm for linear programming solvers.
- Implemented Newton's method as the core iterative approach, deriving iterative equations from the KKT conditions. Extended the framework to a primal-dual approach by reformulating the KKT conditions into residual functions.

Teaching Assistant for School Numerical Analysis course

Feb. 2024–Jun. 2024

- Assistant for lab course and teach basic skills with Matlab.

Technical Skills

Computer Languages: Python, Java, Matlab, R, Lingo, Linux, C++

Developing Tools: NumPy, Scipy, Pandas, PyTorch

刘闻新

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教育背景

华威大学 (University of Warwick)	2026 年 9 月 – 2030 年 9 月
应用数学博士	
帝国理工学院 (Imperial College London)	2024 年 9 月 – 2025 年 11 月
应用数学理学硕士 (科学计算与机器学习方向)	一等优秀
西交利物浦大学 (Xi'an Jiaotong-Liverpool University)	2020 年 9 月 – 2024 年 6 月
应用数学理学学士	一等荣誉

数值分析与优化方向高度相关经历

帝国理工学院研究论文	2025 年 10 月 – 2026 年 5 月
- 论文获欧洲控制会议 2026 (ECC 2026) 录用, 与 Dr. Dante Kalise 合著: http://arxiv.org/abs/2511.14355: 《A PDE-constrained Optimization Approach to Optimal Trajectory Planning under Uncertainty via Reflected Schrödinger Bridges》。 - 论文获数学网络与系统理论国际研讨会 (MTNS) 2026 录用, 与 Dr. Daisuke Inoue 及 Dr. Dante Kalise 合著: 《Computational Methods for Generalized Schrödinger Bridge Problems》。	

国家超算中心科研实习	2025 年 9 月 – 2026 年 3 月
- 开发了用于稀疏线性代数智能求解器选择的 GNN-MLP 混合框架。通过联合学习基于图的结构表示与矩阵级特征, 自动确定最优预条件子与迭代求解器组合。在科学计算应用中大规模稀疏线性系统的求解上实现了超过 3.75 倍的加速。 - 开发了基于机器学习的代数多重网格 (AMG) 求解器优化框架, 实现对强连接阈值 θ 等关键参数的自动调优。提升了复杂物理仿真任务的收敛效率与求解器鲁棒性。 - 通过将初始化问题转化为图表示学习任务, 开发了面向迭代线性求解器的基于 GNN 的初始猜测预测框架。将稀疏矩阵表示为图, 通过图编码器提取结构特征, 并使用基于注意力机制的解码器生成自适应初始解向量。对于具有挑战性的压力泊松系统, 平均求解时间减少 1.79 倍, 最高减少 19.51 倍。	

硕士毕业论文: 最优传输	2025 年 1 月 – 2025 年 9 月
- 采用对称 Douglas-Rachford 分裂法与带近端算子的原-对偶方法求解动态最优传输问题, 并与 Sinkhorn 算法进行性能对比。 - 利用薛定谔桥以动态与静态两种方式近似最优传输问题的解, 实现了基于带近端递推的前向-后向系统的求解算法。 - 实现了带非线性先验漂移与反射边界条件的薛定谔桥问题, 将近端递推算法与有限元方法结合, 求解二维和三维情形下带有障碍物等先验漂移与边界条件的问题。	

计算偏微分方程课程项目	2025 年 1 月 – 2025 年 4 月
- 以显式方法、滞后方法和隐式 Newton 线性化方法求解二阶非线性 ODE 与非线性热方程, 并结合物理意义与理论微扰分析对数值结果进行分析。 - 利用结构化网格映射与自适应网格加密 (AMR) 开发椭圆型 PDE 的数值解, 以提升复杂几何形状下的精度。实现用于大型稀疏线性系统的迭代求解器 (Gauss-Seidel, SOR、多重网格)。 - 求解带 Neumann 与透明边界条件的波动方程, 并通过离散 Fourier 分析评估数值耗散与色散。设计位置相关的完美匹配层 (PML), 将边界反射降低 85%。通过解析 Fourier 分析验证了精度与收敛性。 - 为波-热耦合系统设计 IMEX 有限差分框架, 结合显式波动求解器与隐式热扩散。通过 Fourier 分析与 Jury 准则推导稳定性条件, 并利用交错网格实现高效求解器。通过数值实验展示了热扩散率随变量变化对能量耗散与波动行为的影响。	

有限元方法: 理论与实现课程项目	2025 年 1 月 – 2025 年 4 月
- 以 Python 实现面向对象的有限元框架, 用于求解椭圆型 PDE (Helmholtz/Poisson 方程), 包含 Lagrange 单元、函数空间、结构化网格 (UnitSquareMesh) 与 Gauss 积分规则的模块化类。构建全局节点编号系统以完成局部到全局自由度的映射, 通过逐单元贡献实现高效稀疏矩阵组装, 并利用 einsum 优化的张量运算计算刚度矩阵与质量矩阵。 - 为结构化二维网格上的 Stokes 方程 ($\mu \Delta \mathbf{u} = \nabla p + \mathbf{f}, \nabla \cdot \mathbf{u} = 0$) 构建混合有限元求解器。设计带张量积基函数的矢量值 Lagrange 单元, 通过基于 Vandermonde 矩阵的评估将标量插值扩展至二维速度场。利用 SciPy 稀疏 lil_matrix 格式通过分块矩阵组装 ($A = \begin{bmatrix} A_u & B^T \\ B & 0 \end{bmatrix}$) 实现散度与梯度耦合项, 并通过压力 Poisson 稳定化 (对角扰动) 稳定鞍点系统。使用 einsum 加速应变率张量 ($\epsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$) 的积分运算, 最终将求解器部署用于不可压缩粘性流 ($\mu = 1$) 的模拟, 并以由流函数 ($\psi(x, y) = (1 - \cos 2\pi x)(1 - \cos 2\pi y)$) 导出的解析基准进行验证。	

数据科学课程项目	2025 年 1 月 – 2025 年 4 月
开发用于机场气象条件预测的多分类器 (决策树、随机森林) 与 Huber SVM 模型。通过基于 Gini 重要性的特征选择对随机森林进行优化。以改进的 Huber 损失与 mini-batch SGD 稳定 SVM 训练, 测试精度达 94.1%。	

- 设计带物理约束的复合损失 MLP，用于跑道温度的联合预测。通过冻结预训练 MLP 隐藏层并微调 sigmoid 输出层实现迁移学习用于二分类，精度达 99.8%，优于无迁移学习模型的 97%。
- 在 PyTorch 中设计 U-Net 卷积神经网络，用于联合降维与基因表达回归。利用早停与 Dropout 在回归任务上取得 $R^2 = 0.69$ 。
- 利用归一化 Laplacian 特征向量构建谱布局，用于数据分析与聚类，并通过二部直方图进行度中心性分析。

计算线性代数课程项目

2024 年 9 月 – 2025 年 1 月

- 开发基于 QR 分解的最小二乘求解器（超定/欠定系统），实现四种算法：法方程、经典/改进 Gram-Schmidt 及 Householder (Python/NumPy)。针对病态最小二乘问题设计基于增广 QR 分解的正则化方法，以 $\lambda = 10^{-6}$ 惩罚项将条件数降低 5 个数量级。
- 实现并分析用于计算实对称矩阵特征值的实用 QR 算法，包括 Wilkinson 位移与压缩等改进。通过对不同矩阵规模的对比实验展示 Wilkinson 位移加速收敛的效率，并通过反迭代验证特征值精度。
- 通过实现 Jacobi 与 Gauss-Seidel 预条件子对 GMRES 算法进行扩展，以改善病态系统的收敛性。对不同 ϵ 值下的残差衰减与收敛速率进行对比分析，表明预条件子能显著提升性能，在 Gauss-Seidel 预条件子下随病态程度增加效果尤为突出。

科学计算课程项目

2024 年 9 月 – 2025 年 1 月

- 实现并分析线性搜索、带二分查找的归并排序 ($O((m+n)\log n)$)，用于大规模数据集的搜索与模式匹配。将 Rabin-Karp 算法应用于氨基酸序列的模式匹配，利用优化的哈希计算实现快速序列对比。
- 开发并分析非线性 ODE 动力学系统的数值解。实现了微扰能量计算与仿真对比的测试，验证了一致性与精度。对不同系统规模下的周期、非周期与混沌行为进行分析，利用主成分分析 (PCA) 与关联维数分析识别混沌动力学。
- 为 Lorenz 系统实现扩展 Kalman 滤波器，分析其单步计算代价。重点识别方法中计算量最大的部分，并探究随问题规模变化的墙时钟时间趋势。

本科毕业论文：数值优化

2023 年 9 月 – 2024 年 6 月

- 通过利用指示函数将不等式约束问题转化为等式约束形式，应用内点法进行数值近似求解，并为线性规划求解器实现该算法。
- 以 Newton 法作为核心迭代方法，从 KKT 条件推导迭代方程。通过将 KKT 条件改写为残差函数，将框架扩展为原-对偶方法。

数值分析课程助教

2024 年 2 月 – 2024 年 6 月

- 协助实验课教学，讲授 MATLAB 基础操作技能。

技术技能

编程语言：Python、Java、Matlab、R、Lingo、Linux、C++

开发工具：NumPy、Scipy、Pandas、PyTorch